

Exercise 5.15

Since A is orthogonal, assume λ is an eigenvalue of A , then

$$Ax = \lambda x$$

for some x .

Then

$$\|Ax\| = \|\lambda x\| = \|x\|$$

by definition of orthogonal matrix.

Then

$$\|\lambda x\| = |\lambda| \|x\| = \|x\|$$

implies

$$|\lambda| = 1 \Rightarrow \lambda \in \{-1, 1\}.$$

So the only possible eigenvalues are 1 and -1.

Since A is diagonalizable, then there exists a matrix P such that P is invertible, and

$$A = P D P^{-1}.$$

As we know that $\det(A)=1$, hence

$$\begin{aligned} \det(A) &= \det(P D P^{-1}) \\ &= \det(P) \det(D) \det(P^{-1}) \\ &= \det(P) \det(D) \frac{1}{\det(P)} = \det(D). \end{aligned}$$

And since

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix},$$

where λ_1, λ_2 are eigenvalues of A ,

$$\det(A) = \det(D) = \lambda_1 \lambda_2 = 1.$$

Since $\lambda_1, \lambda_2 \in \{-1, 1\}$ and $\lambda_1 \lambda_2 = 1$, then

$$D = I \quad \text{or} \quad D = -I.$$

If $D = I$, then

$$A = P I P^{-1} = P P^{-1} = I.$$

If $D = -I$, then

$$A = P(-I)P^{-1} = -P P^{-1} = -I.$$

Exercise 5.6

Since L is an isomorphism, L^{-1} exists.

Then consider

$$M \circ L = L^{-1} \circ (L \circ M) \circ L.$$

Then, let's compute the characteristic polynomial of $M \circ L$.

$$\begin{aligned} \det(M \circ L - \lambda I) &= \det(L^{-1} \circ (L \circ M) \circ L - \lambda I) \\ &= \det(L^{-1} \circ (L \circ M - \lambda I) \circ L) \\ &= \det(L^{-1}) \det(L \circ M - \lambda I) \det(L). \end{aligned}$$

Since L is invertible,

$$\begin{aligned} &= 1/\det(L) \det(L \circ M - \lambda I) \det(L) \\ &= \det(L \circ M - \lambda I). \end{aligned}$$

Hence

$$\det(L \circ M - \lambda I) = \det(M \circ L - \lambda I).$$

Exercise 5.8

Let

$$A = \begin{bmatrix} p & -q \\ q & p \end{bmatrix}, \quad p, q \in \mathbb{R}, \quad q \neq 0.$$

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} p-\lambda & -q \\ q & p-\lambda \end{bmatrix} \\ &= (p-\lambda)^2 + q^2 > 0 \end{aligned}$$

over $\mathbb{R}$, since $q \neq 0$.

So there are no real eigenvalues over $\mathbb{R}$.

Over $\mathbb{C}$,

$$\begin{aligned} (p-\lambda)^2 + q^2 &= (p-\lambda-qi)(p-\lambda+qi) \\ &= ((p-qi)-\lambda)((p+qi)-\lambda). \end{aligned}$$

Hence

$$\lambda_1 = p - qi, \quad \lambda_2 = p + qi.$$

Then A has two distinct eigenvalues, so it is diagonalizable over $\mathbb{C}$.

For $\lambda_1 = p - qi$,

$$Ax = (p-qi)x$$

means

$$\begin{bmatrix} p & -q \\ q & p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (p-qi) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

So

$$\begin{bmatrix} p x_1 - q x_2 \\ q x_1 + p x_2 \end{bmatrix} = \begin{bmatrix} (p-qi)x_1 \\ (p-qi)x_2 \end{bmatrix}.$$

Thus

$$q x_2 = q i x_1 \Rightarrow x_2 = i x_1,$$

and

$$q x_1 = -q i x_2 \Rightarrow x_1 = -i x_2.$$

Therefore

$$V^A_{\{\lambda_1\}} = \text{span}\{ (1, i)^T \}.$$

For $\lambda_2 = p + qi$,

$$Ax = (p+qi)x$$

gives us

$$x_2 = -i x_1, \quad x_1 = i x_2.$$

Therefore

$$V^A_{\{\lambda_2\}} = \text{span}\{ (1, -i)^T \}.$$

Exercise 5.19

(i)

Let λ in $\mathbb{C}$ be an eigenvalue of L . This means

$$L(x_0, x_1, x_2, \dots) = \lambda(x_0, x_1, x_2, \dots),$$

which means

$$(x_1, x_2, \dots) = (\lambda x_0, \lambda x_1, \dots).$$

Therefore

$$x_{i+1} = \lambda x_i$$

for all $i = 0, \dots, \infty$.

And this is exactly the definition of a complex infinite geometric sequence.

And surely, for every λ in $\mathbb{C}$, there exists a geometric sequence such that

$$x_{i+1} = \lambda x_i.$$

Hence every λ in $\mathbb{C}$ is an eigenvalue of L .

The eigenspace V^L_λ associated with λ is all sequences $(x_n)_{n=0}^\infty$ such that

$$x_{i+1} = \lambda x_i$$

for all $i = 0, \dots, \infty$.

If $x_0 \neq 0$, then

$$x_n = \lambda^n x_0.$$

So

$$V^L_\lambda = \text{span}\{(1, \lambda, \lambda^2, \dots)\}.$$

(ii)

$$R(x_0, x_1, \dots) = (0, x_0, x_1, \dots).$$

If λ is an eigenvalue of R , then

$$\begin{aligned} R(x_0, x_1, \dots) &= \lambda(x_0, x_1, \dots) \\ &= (0, x_0, x_1, \dots). \end{aligned}$$

Entrywise,

$$\begin{aligned} \lambda x_0 &= 0, \\ \lambda x_1 &= x_0, \\ \lambda x_2 &= x_1. \end{aligned}$$

So

$$\lambda x_i = x_{i-1}$$

for all $i = 1, \dots, \infty$.

Let $x_{-1} = 0$.

Then let's consider two cases.

Case 1: $\lambda = 0$. Then

$$x = (0, \dots, 0),$$

which contradicts that λ is an eigenvalue of R .

Case 2: $\lambda \neq 0$.

Then

$$\lambda x_0 = 0 \Rightarrow x_0 = 0.$$

Consequently,

$$\lambda x_1 = 0 \Rightarrow x_1 = 0,$$

and generally

$$\lambda x_i = 0$$

for all $i = 0, \dots, \infty$.

Hence

$$x_i = 0$$

for all $i = 0, \dots, \infty$, which contradicts that x is an eigenvector.

Therefore, by two cases,

$$\sigma(R) = \text{empty set}.$$